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RESEARCH ARTICLE

Quinoxaline-Based Single-Component Ambipolar Polymer for Organic Electrochemical Transistors

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ABSTRACT

Ambipolar organic mixed ion-electron conductors (OMIECs) are essential for simplifying the fabrication of organic electrochemical transistor (OECT)-based complementary logic circuits and bioelectronics. However, the limited availability of ambipolar OMIEC materials restricts their further advancement. By incorporating a thiadiazol ring fusion onto the quinoxaline building block and introducing difluorobenzothiadiazole as a secondary acceptor, we successfully synthesized a novel ambipolar OMIEC named TdQ-DT2FBT. Due to the strong electron-withdrawing capability of the [1,2,5]thiadiazolo[3,4-g]quinoxaline (TdQ) unit, TdQ-DT2FBT has achieved a narrow bandgap of 1.16 eV. TdQ-DT2FBT demonstrates a high product of mobility and capacitance (μC^*) of 301.03 F cm^{-1} V^{-1} s^{-1} for the p-type mode and a respectable μC^* of 11.27 F cm^{-1} V^{-1} s^{-1} for the n-type mode. Vertical OECTs were fabricated, achieving high and relatively balanced ambipolar properties with transconductance values of 211.50 mS for the p-type mode and 79.77 mS for the n-type mode. Furthermore, a vertical complementary inverter was developed, achieving a gain of 78 V V^{-1} at $V_{dd} = 1.0$ V, and a voltage amplification gain of approximately 15 for electrooculogram recording. This work presents a novel quinoxaline-based single-component organic mixed ionic-electronic conductor (OMIEC), paving the way for the fabrication of high-performance ambipolar OECTs and related complementary circuits.

1 | Introduction

Organic electrochemical transistors (OECTs) have garnered significant attention as a powerful platform for wearable electronics [1–4], bioelectronics [5–7], physiological sensors [8, 9], health monitoring [10, 11], and neuromorphic devices [12, 13]. This attention stems from their unique operating mechanism, high transconductance, low operating voltage (< 1 V), and good biocompatibility—key advantages that distinguish them from traditional electronic devices [14–18]. Unlike conventional field-effect transistors (FETs), which rely on the capacitive accumulation or depletion of charge carriers at the dielectric/active layer interface, OECTs operate through bulk doping and dedoping of organic channel materials via electrochemical reactions [19, 20]. Organic mixed ionic-electronic conductors (OMIECs), materials that enable simultaneous ionic and electronic charge transport, are the most promising channel materials for OECTs. In recent years, significant progress has been made in developing

Songlin Mo and Ting Wang contributed equally to this work.

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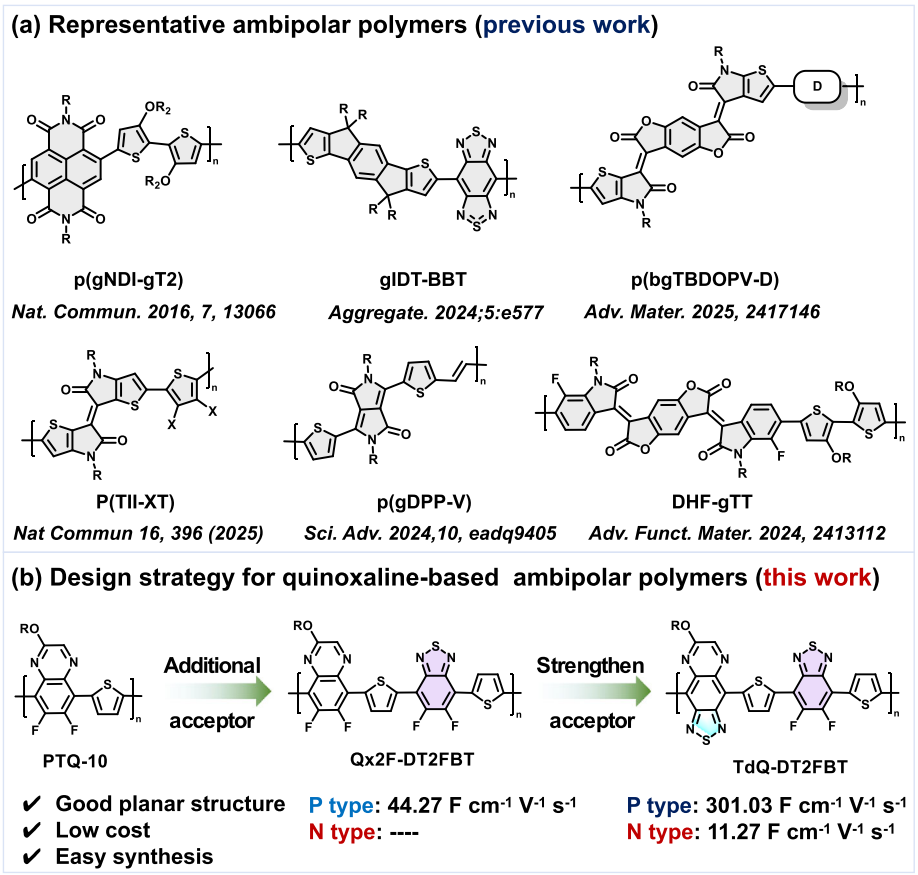


FIGURE 1 | (a). Chemical structures of representative ambipolar OMIECs in reported literature. (b). Design strategy for quinoxaline-based ambipolar polymers.

high-performance p-type (hole-transporting) OMIECs, driven by continuous material innovation and device optimization [21, 22]. By contrast, the development of n-type (electron-transporting) OMIECs has lagged relatively far behind their p-type counterparts [23–26]. This imbalance in the performance of p-type and n-type OMIECs severely hinders the development of OECT-based complementary logic circuits (e.g., NOT, NAND, and NOR gates), which require both carrier types to function. Single-component ambipolar OMIECs have emerged as an alternative solution for constructing ambipolar OECTs and complementary logic circuits, as they offer the advantages of simplifying the fabrication process and lowering costs [27–30]. However, only a limited number of ambipolar OMIECs classes have been reported to date, highlighting an unmet need in the field.

Ideal ambipolar OMIECs require two critical features: narrow bandgaps and well-aligned frontier orbital energy levels. These characteristics are essential for facilitating high and balanced electron/hole mobilities [30]. To achieve these goals, researchers have incorporated strong electron-deficient building blocks, such as naphthalene diimide (NDI) [31–33], diketopyrrolopyrrole (DPP) [28, 30, 34–36], bisistain-lactone (BIL) [27], thiophene-fused benzodifurandione-based oligo(*p*-phenylenevinylene) (TBDOPV) [37], benzobisthiadiazole (BBT) [38], and thienoisoindigo (TII) [39] into ambipolar OMIEC polymers, as shown in Figure 1a. For instance, Yue et al. reported a single-component ambipolar polymer, p(gDPP-V), by combining the DPP acceptor with the weak electron-donating vinylene unit. This material exhibits a product of mobility and capacitance (μC^*) of 204 $F\,cm^{-1}\,V^{-1}\,s^{-1}$ in p-type mode and a μC^* of 102 $F\,cm^{-1}\,V^{-1}\,s^{-1}$ in n-type mode. The resulting highly stable vertical OECT-based inverter shows a high voltage gain of 105 $V\,V^{-1}$ under a low driving voltage of 0.8 V [28]. In another study, Lei et al. designed a high-spin polymer, P(TII-2FT), which exhibits highly stable and balanced ambipolar properties, with a p-type μC^* of 158.6 $F\,cm^{-1}\,V^{-1}\,s^{-1}$ and an n-type μC^* of 147.4 $F\,cm^{-1}\,V^{-1}\,s^{-1}$. Devices based on P(TII-2FT) achieved remarkable voltage gains while maintaining a compact form factor. More importantly, this amplifier enabled on-site capture and amplification of diverse electrophysiological signals (including human electrocardiogram (ECG), electroencephalogram (EEG), and mouse electrocorticogram (ECoG)) and delivered a higher signal-to-noise ratio (SNR) than conventional electrodes [39]. However, most of the aforementioned strong electron-deficient building blocks face significant synthetic challenges, typically requiring multiple reaction steps and suffering from low overall yields.

In contrast to many conventional acceptor units such as NDI, DPP, and TII that often require complicated multistep syntheses, quinoxaline (Qx) derivatives offer a promising alternative to these complex building blocks, as they exhibit relatively simple structures, straightforward synthetic routes, and high overall yields [40]. Additionally, the highly planar structure of Qx facilitates efficient charge delocalization along the conjugated chain, thereby enhancing the carrier mobility of

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Qx-based materials [41]. Meanwhile, the chemically robust heteroaromatic framework exhibits excellent stability under repeated redox conditions. Furthermore, the multiple reactive positions within the Qx segment allow flexible molecular engineering, where electron-withdrawing groups (EWGs) can be strategically introduced to finely tune the LUMO energy level and enhance electron-transport capability. Despite their extensive application in organic field-effect transistors (OFETs) and organic photovoltaics (OPVs) [42–44], Qx-based OMIECs have not been reported to date—representing a critical research gap.

Inspired by the cost-effectiveness and synthetic accessibility of Qx-based materials (e.g., PTQ10) [44], we aimed to develop Qx-based ambipolar OMIECs. Given the weak electron-withdrawing ability of the parent Qx moiety, additional electron-withdrawing units/groups are necessary to achieve both a low-lying LUMO energy level and a narrow bandgap—two prerequisites for ambipolar behavior. To address this, we pursued two synthetic strategies: 1. Incorporating difluorobenzothiadiazole (2FBT) as a second acceptor unit to construct Acceptor1-Donor-Acceptor2 (A1-D-A2)-type polymers; 2. Fusing a thiadiazole ring onto the Qx skeleton to form a stronger electron-withdrawing building block, [1,2,5]thiadiazolo[3,4-g]quinoxaline (TdQ), which was then integrated into A1-D-A2-type polymers. Through these strategies, we designed and synthesized two polymers: Qx2F-DT2FBT and TdQ-DT2FBT (Figure 1b). Compared to traditional donor-acceptor (D-A)-type polymer semiconductors, these dual-acceptor polymers exhibit greater tunability in intermolecular interactions and energy level regulation [45, 46]. The A1-D-A2 configuration also enhances intramolecular charge transfer (ICT), enables precise energy level tuning, maintains molecular planarity, and promotes π-π stacking in the solid state—all of which are critical for achieving efficient charge transport and high-performance OECTs. Our results indicate that TdQ-DT2FBT exhibits stronger ICT and a narrower bandgap than Qx2F-DT2FBT, enabling reversible electrochemical doping under both p-type and n-type operating conditions. In addition, TdQ-DT2FBT demonstrates clear ambipolar behavior, with a high μC^* of 301.03 $F{\cdot}cm^{-1}{\cdot}V^{-1}{\cdot}S^{-1}$ in p-type mode and 11.27 $F\ cm^{-1}\ V^{-1}\ s^{-1}$ in n-type mode. In contrast, Qx2F-DT2FBT functions only in p-type mode, with a low μC^* of 44.27 $F\ cm^{-1}\ V^{-1}\ s^{-1}$. Furthermore, TdQ-DT2FBT-based vertical OECTs achieved relatively balanced ambipolar characteristics, including a maximum transconductance (g_m) of 211.50 mS in p-type mode and 79.77 mS in n-type mode, along with rapid response times (~tens of milliseconds) and robust cycling stability (~1000 s). Leveraging the ambipolar properties, we developed a vertical inverter that achieved a high voltage gain of 78 $V\ V^{-1}$ at a supply voltage (V_{dd}) of 1.0 V. This inverter was subsequently applied successfully for the amplification of electrooculogram (EOG) signals. These results demonstrate that rational chemical modification of the quinoxaline building block is an effective strategy for obtaining single-component ambipolar OMIECs materials that combine straightforward synthesis with high OECT performance. This work not only fills the research gap of Qx-based OMIECs but also provides a versatile design principle for developing high-performance ambipolar materials for next-generation bioelectronic and logic devices.

2 | Results and Discussion

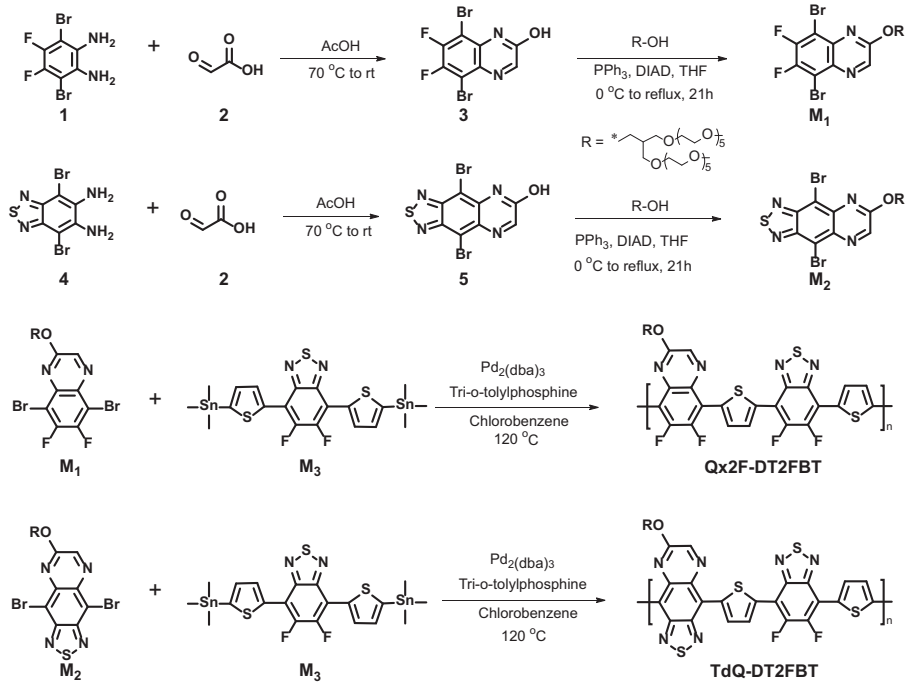
2.1 | Polymer Synthesis and Characterization

The synthetic routes for Qx2F-DT2FBT and TdQ-DT2FBT are illustrated in Scheme 1, with detailed procedures provided in the Supporting Information. The Qx and TdQ units can be synthesized through a high-yield one-step ring-closing reaction; however, their rigid molecular structures lead to significant intermolecular interactions, resulting in poor solubility. To address this challenge, hydrophilic ethylene glycol (EG) side chains were incorporated into the polymer backbone—this modification not only enhances solubility but also shows potential for improving ion mobility. Consequently, Qx and TdQ acted as nucleophiles in S_N2 reactions, facilitating the introduction of EG side chains to yield two distinct monomers (M_1 and M_2). These monomers subsequently underwent palladium-catalyzed Stille coupling with 5,6-difluoro-4,7-bis(5-(trimethylstannyl)thiophen-2-yl)benzo[c][1,2,5]thiadiazole (M_3) to produce the target polymers Qx2F-DT2FBT and TdQ-DT2FBT. The purification of these polymers was achieved via Soxhlet extraction using methanol, acetone, and hexane; they were ultimately recovered from the chloroform fraction. The number-average molecular weights (M_n) and polydispersity indices (PDIs) of the polymers were determined by gel permeation chromatography (GPC) using chloroform as the eluent. Qx2F-DT2FBT exhibited M_n = 22.2 kDa with a PDI of 1.87, while TdQ-DT2FBT showed M_n = 21.0 kDa with a PDI of 3.77 (Figure S1).

2.2 | Theoretical Calculations

Density functional theory (DFT) calculations were performed at the B3LYP/6-31G(d) level for three repeating units of PTQ10, Qx2F-DT2FBT, and TdQ-DT2FBT polymers to comprehensively analyze their electronic properties and geometric configurations (Figure 2a). For structural optimization purposes, EG side chains of identical length were replaced by methoxy groups. Notably, all three polymers exhibit nearly planar conformations, and this planarity within their main chain structures is expected to enhance π-π interactions between adjacent chains, which holds significant importance for intramolecular charge transfer and intermolecular charge hopping [28, 47]. The highest occupied molecular orbital (HOMO) and lowest unoccupied molecular orbital (LUMO) wavefunctions for all polymers exhibit delocalized characteristics along the conjugated main chain (Figure 2b). For PTQ10, the HOMO and LUMO energy levels are −5.15 and −2.49 eV, respectively, resulting in a band gap of 2.66 eV. In comparison to PTQ10, the introduction of a second acceptor unit in Qx2F -DT2FBT significantly lowers the LUMO energy level to −3.02 eV, resulting in a reduced bandgap of 2.04 eV. Upon introducing the stronger electron-accepting unit TdQ into TdQ-DT2FBT, both the LUMO energy level and bandgap are further reduced to −3.37 and 1.42 eV, respectively. The deeper LUMO energy level and narrower bandgap highlight TdQ-DT2FBT's potential as ambipolar materials.

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SCHEME 1 | Synthetic route to the polymers.

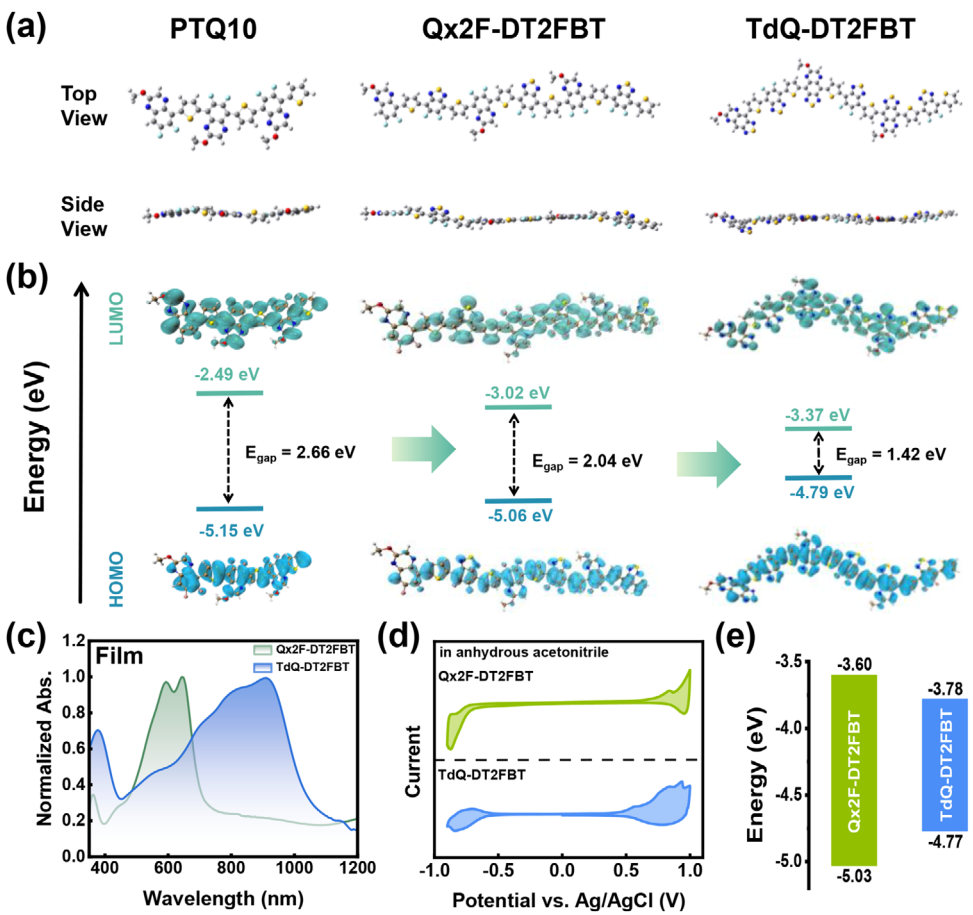


FIGURE 2 | (a) The results of structural optimization for the trimer structure of PTQ10, Qx2F-DT2FBT and TdQ-DT2FBT. (b) The FMO energy levels of PTQ10, Qx2F-DT2FBT, and TdQ-DT2FBT. Calculations are carried out at the B3LYP/6-31G(d) level. (c) Absorption spectra for Qx2F-DT2FBT and TdQ-DT2FBT in the thin-film state. (d) Cyclic voltammetry curves for the films of Qx2F-DT2FBT and TdQ-DT2FBT measured in anhydrous acetonitrile solution containing 0.1 M tetrabutylammonium hexafluorophosphate (Bu_4NPF_6). (e) Energy levels diagrams of Qx2F-DT2FBT and TdQ-DT2FBT.

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TABLE 1 | Summaries of absorption and electrochemical properties of OMIECs.

Polymer	λ_{max}^{sol} (nm)	λ_{max}^{film} (nm)	λ_{onset}^{film} (nm)[a]	$E_{red, org}$ (V)	$E_{ox, org}$ (V)	$E_{HOMO, CV}$ (eV)[b]	$E_{LUMO, CV}$ (eV)[b]	$E_{g, opt}$ (eV)[c]
Qx2F-DT2FBT	636	645	720	−0.72	0.71	−5.03	−3.60	1.72
TdQ-DT2FBT	878	912	1072	−0.54	0.45	−4.77	−3.78	1.16

[a]UV-visible absorption onset in thin films.

[b]E_{LUMO} and E_{HOMO} were calculated from the equation $E_{LUMO} = -(4.8 - E_{Fc/Fc^+} + E_{red, org})$ and $E_{HOMO} = -(4.8 - E_{Fc/Fc^+} + E_{ox, org})$, where the E_{Fc/Fc^+} is +0.48 V.

[c]Calculated from the absorption band edge of the polymer film, $E_{g, opt} = 1240/\lambda_{onset}^{film}$.

2.3 | Photophysical and Electrochemical Properties

To investigate the optical properties of Qx2F-DT2FBT and TdQ-DT2FBT, we measured the UV–vis absorption spectra of both polymers, as illustrated in Figure 2c; Figure S2. The relevant data are summarized in Table 1. Both polymers exhibit similar spectral absorption characteristics in solution and thin films, displaying distinct high-energy and low-energy absorption bands. The high-energy absorption bands observed within the wavelength range of 340–400 nm correspond to π-π^* transitions, while the low-energy absorption bands appearing between 400–1 100 nm are attributed to intramolecular charge transfer (ICT) between donors and acceptors and aggregation-induced absorption. The ICT absorption maximum of Qx2F-DT2FBT in solution is located at 636 nm, with a slight red shift to 645 nm in the film spectra. In contrast, TdQ-DT2FBT exhibits an ICT absorption maximum at 878 nm in the solution spectrum, which shows a pronounced red shift to 912 nm in the film spectrum. In addition, the optical bandgap, estimated from the onset of film absorption, is calculated to be 1.72 eV for Qx2F-DT2FBT and 1.16 eV for TdQ-DT2FBT. This significant redshift in the absorption peak, along with a narrower bandgap, indicates enhanced intermolecular interactions in the solid-state form of TdQ-DT2FBT [36].

The electrochemical properties of Qx2F-DT2FBT and TdQ-DT2FBT were evaluated by cyclic voltammetry (CV) in anhydrous acetonitrile solution containing 0.1 M tetrabutylammonium hexafluorophosphate (Bu_4NPF_6), with ferrocene employed as an external standard (Figure S3). The results are summarized in Table 1. As shown in Figure 2d,e, the reduction potential ($E_{red, org}$) and oxidation potential ($E_{ox, org}$) for Qx2F-DT2FBT were measured to be −0.72 and 0.71 V, respectively, while those for TdQ-DT2FBT were −0.54 and 0.45 V. Consequently, the HOMO/LUMO energy levels of Qx2F-DT2FBT and TdQ-DT2FBT are determined to be −5.03 eV/−3.60 eV and −4.77 eV/−3.78 eV, respectively. Overall, the CV results indicate that TdQ-DT2FBT exhibits a deeper LUMO energy level and a shallower HOMO energy level, as well as a narrower bandgap. These findings are in strong agreement with DFT calculations and can be primarily attributed to the enhanced electron-withdrawing capability of the TdQ unit.

Spectroelectrochemical (SEC) measurements further elucidated the doping mechanism of Qx2F-DT2FBT and TdQ-DT2FBT films under varying bias voltages in a 0.1 M NaCl aqueous solution. Qx2F-DT2FBT exhibits minimal polaronic/bipolaronic formation under negative bias and virtually none under positive bias (Figure 3a,b,e,f). In contrast, when positive bias voltages are applied from 0 to 1.0 V and negative bias voltages from 0 to −1.0 V, the intensity of the main ICT absorption peak (912 nm) in TdQ-DT2FBT significantly decreased, while new absorption bands emerged within the wavelengths of 440–650 and 950–1400 nm. This indicates the formation of polaronic/bipolaronic species within the polymer (Figure 3c,d,g,h). For quantitative evaluation of electrochemical doping behavior, the polaron-to-ICT absorption intensity ratio was determined. As illustrated in Figure S4, TdQ-DT2FBT possesses a significantly higher ratio than Qx2F-DT2FBT under equivalent bias conditions, indicating more efficient polaron production in the doping process. These findings indicate that TdQ-DT2FBT can be readily p-doped or n-doped, in contrast to the unipolar p-type Qx2F-DT2FBT. This characteristic presents the potential for achieving ambipolar charge transport.

2.4 | OECT Device Fabrication and Characterization

The planar structure OECT devices based on Qx2F-DT2FBT and TdQ-DT2FBT were fabricated, as detailed in Figure S5 and the experimental section. The corresponding OECT parameters are summarized in Table 2, where the reported values represent averages obtained from at least three independently fabricated devices, demonstrating good device-to-device reproducibility. The typical transfer curves and output characteristics presented in Figures S6–S8 indicate that Qx2F-DT2FBT operates exclusively under negative gate voltages, whereas TdQ-DT2FBT functions under both positive and negative gate voltages, exhibiting ambipolar behavior. In p-type mode, Qx2F-DT2FBT achieves a maximum transconductance (g_m) of 0.55 ± 0.03 mS at a drain voltage (V_{DS}) of −0.60 V and a gate voltage (V_{GS}) of −1.0 V. In contrast, TdQ-DT2FBT exhibits a significantly higher maximum g_m of 7.55 ± 0.35 mS at V_{DS} = −0.6 V and V_{GS} = −1.0 V. Furthermore, TdQ-DT2FBT exhibits a significantly lower threshold voltage (V_{TH}) of −0.40 V compared to the V_{TH} of −0.76 V for Qx2F-DT2FBT (Figure S7), indicating its capability to operate at relatively low voltages consistent with its shallower HOMO energy level. In n-type mode, Qx2F-DT2FBT exhibits no n-type characteristics; however, TdQ-DT2FBT demonstrates a maximum g_m of 0.18 ± 0.02 mS at V_{DS} = 0.6 V and V_{GS} = 1.0 V.

Additionally, we evaluated the μC^* values of the two materials based on the equation: $g_m = \frac{\partial I_D}{\partial V_G} = \frac{Wd}{L} \times \mu C^* \times |V_G - V_{TH}|$.

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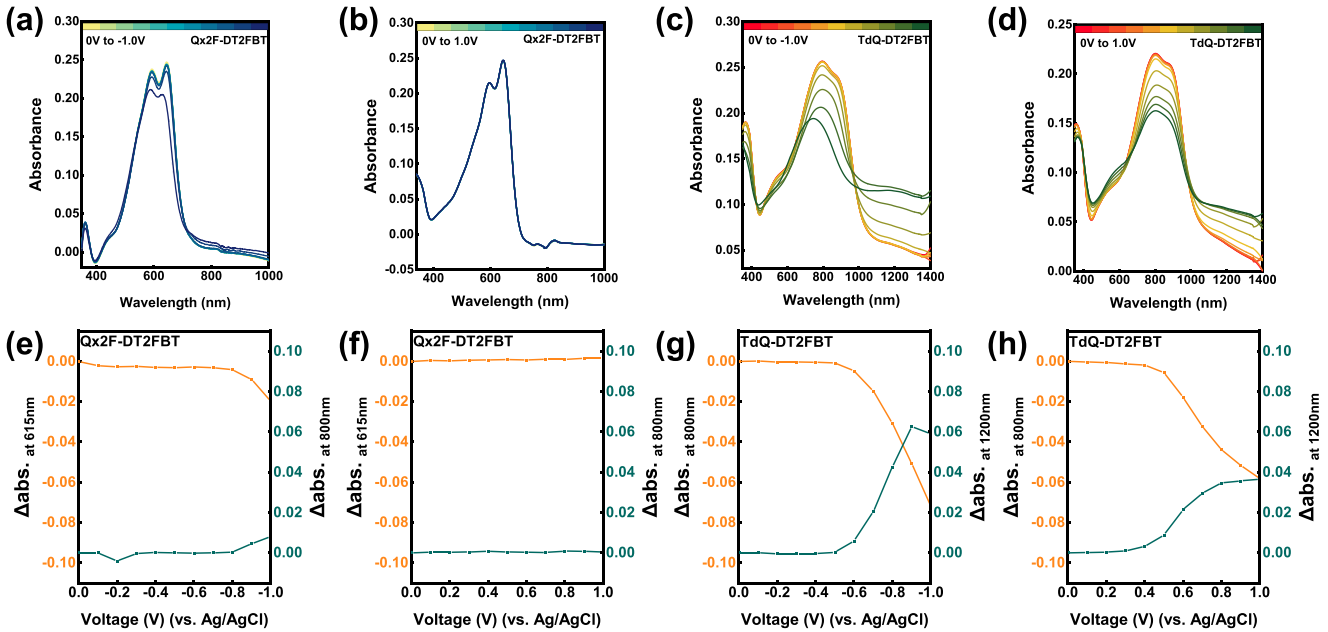


FIGURE 3 | (a–d) Electrochemical absorption spectra of Qx2F-DT2FBT and TdQ-DT2FBT films in a 0.1 M NaCl aqueous solution, with varying bias voltages applied relative to Ag/AgCl reference electrode. (e–h) Variations in absorption intensity at designated wavelengths within the electrochemical absorption spectrum.

TABLE 2 | Summary of planar OECTs' Performance.

Polymer	d[a] [nm]	g_m[b] [mS]	$g_{m,norm}$[c] [mS]	V_{TH}[d] [V]	μC^*[e] [F cm^{-1} V^{-1} s^{-1}]	C^*[f] [F cm^{-3}]	μ^g [cm^2 V^{-1} s^{-1}]	I_{on}/I_{off}
TdQ-DT2FBT(p)	27.4 ± 0.3	7.55 ± 0.35	165.35 ± 7.71	−0.40	301.03 ± 14.10	351.37 ± 16.45	0.857	10^5
TdQ-DT2FBT(n)	27.4 ± 0.3	0.18 ± 0.02	3.83 ± 0.47	0.66	11.27 ± 1.36	245.30 ± 10.32	0.043	10^3
Qx2F-DT2FBT(p)	31.3 ± 0.4	0.55 ± 0.03	10.65 ± 0.59	−0.76	44.27 ± 2.44	184.67 ± 24.44	0.240	10^3

[a]Average data extracted from a step profilometer.
[b]The average value was taken after testing both polymers in devices with identical channel size. (W/L = 500/30 μm).
[c]Normalized by channel geometry.
[d]V_{th} was determined by extrapolating the corresponding $I_{DS}^{1/2}$ versus V_{GS} plot.
[e]The μC^* was calculated according to the equation: $g_m = \frac{\partial I_D}{\partial V_G} = \frac{Wd}{L} \times \mu C^* \times |V_G - V_{TH}|$, where W, L, and d represent the channel width, length, and film thickness, respectively. V_{GS} and V_{TH} represent the applied gate voltage and threshold voltage, respectively.
[f]The average C^* value determined from EIS measurements.
[g]Calculated from the figure of (μC^*) and volumetric capacitance (C^*).

Qx2F-DT2FBT exhibited a maximum μC^* of 46.71 F cm^{-1} V^{-1} s^{-1} in p-type mode. In contrast, TdQ-DT2FBT demonstrated a maximum μC^* value of 315.13 F cm^{-1} V^{-1} s^{-1} in p-type mode and a respectable μC^* value of 12.63 F cm^{-1} V^{-1} s^{-1} in n-type mode. For an in-depth elucidation of performance discrepancies, volumetric capacitance (C^*), a key parameter governing OECT performance that quantifies the ionic charge storage capacity of the active layer post-electrochemical doping, was characterized for both polymers via electrochemical impedance spectroscopy (EIS), as shown in Figure S9. In p-type operation, Qx2F-DT2FBT achieved a C^* value of 211.11 F·cm^{-3}, whereas TdQ-DT2FBT attained a higher C^* value of 367.82 F·cm^{-3}; TdQ-DT2FBT also demonstrated a high C^* value of 255.62 F·cm^{-3} in n-type mode. The reduced C^* value observed in Qx2F-DT2FBT can be attributed to its relatively enhanced hydrophobic properties, as confirmed by contact angle measurements. As shown in Figure S10, the contact angle increased from 74° for TdQ-DT2FBT to 87° for Qx2F-DT2FBT. This improvement may stem from the higher crystallinity of the Qx2F-DT2FBT film (see below). Hole and electron mobilities (μ_h and μ_e) of the OECTs were derived from μC^* and C^* values: TdQ-DT2FBT delivered μ_h = 0.857 cm^2 V^{-1} s^{-1} and μ_e = 0.043 cm^2 V^{-1} s^{-1}, while Qx2F-DT2FBT showed a lower μ_h = 0.240 cm^2 V^{-1} s^{-1}.

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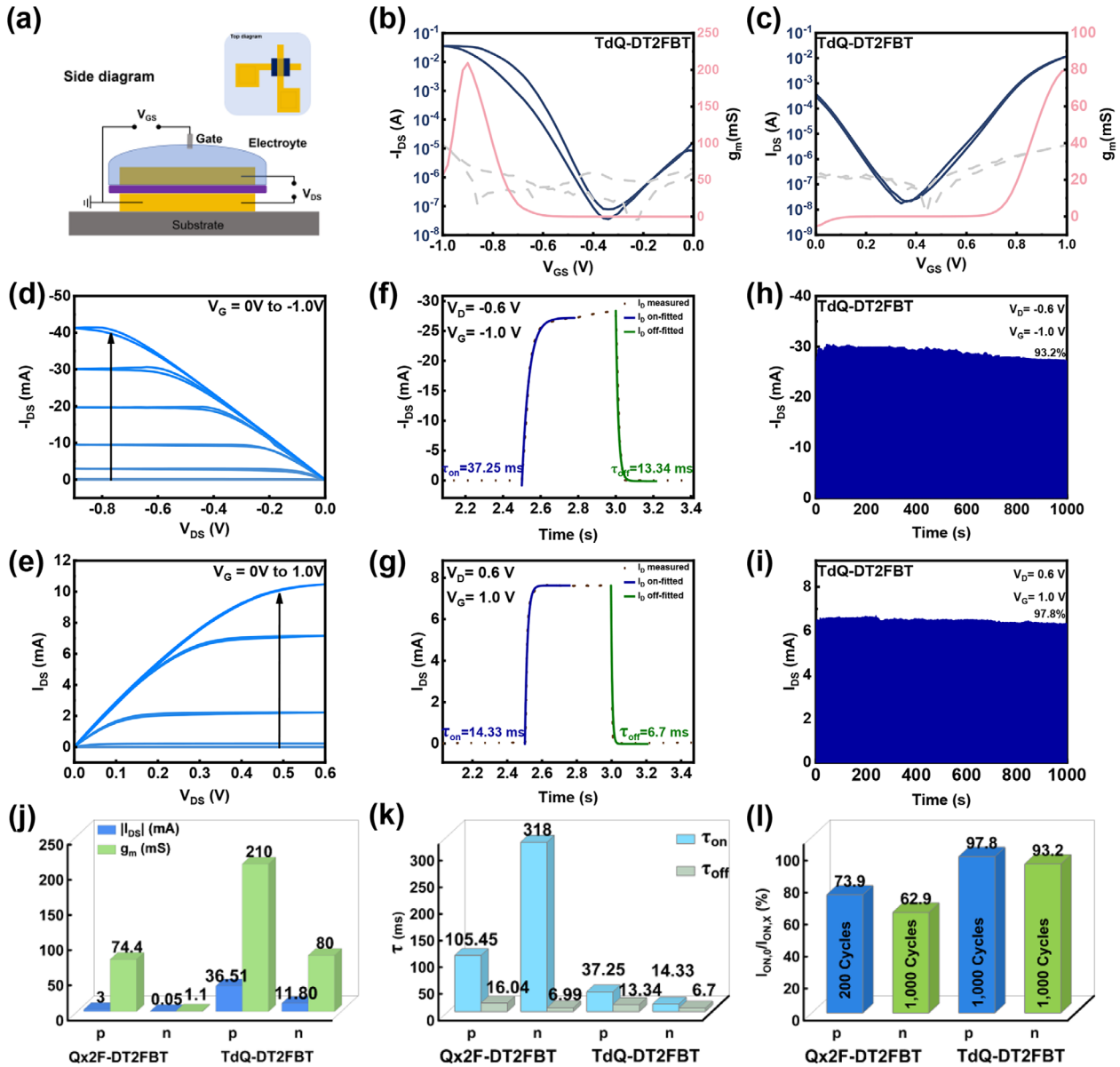


FIGURE 4 | (a) Schematic diagram of vertical organic electrochemical transistor (OECT) device structure. (b, c) Typical OECT transfer curves (I_D: blue lines; I_G: gray dotted lines) and g_m curves (pink lines) of vOECTs-based TdQ-DT2FBT:DtFDA (6:1) with NaCl (0.1 M) as electrolyte. (d, e) Output curves of vOECTs-based TdQ-DT2FBT:DtFDA (6:1) with NaCl (0.1 M) as electrolyte. (f, g) Transient profiles of vOECTs-based TdQ-DT2FBT:DtFDA (6:1) with NaCl (0.1 M) as electrolyte. (h, i) Operational stability of vOECTs-based TdQ-DT2FBT:DtFDA (6:1) with NaCl (0.1 M) as electrolyte. (j) The maximum $|I_D|$ and g_m. (k) τ_{on}/τ_{off} response times. (l) The performance of Qx2F-DT2FBT:DtFDA (6:1) and TdQ-DT2FBT:DtFDA (6:1) deteriorates after prolonged on-off switching.

Previous studies have demonstrated that vertical OECT (vOECT) not only significantly enhances performance (especially for n-type polymers), but also enables smaller effective channel dimensions [48]. Therefore, we introduce the vOECT structure (Figure 4a) to further investigate the properties of the two polymers, with the aim of achieving a more balanced n-type and p-type performance. The detailed fabrication process of the vOECTs is shown in Figure S11. Notably, we incorporated the crosslinker double-end trifluoromethyl diazirine (DtFDA) into OMIEC [49], a strategy that enables precise control over channel dimensions through patterning and enhances structural stability via UV-induced chemical crosslinking. This approach effectively suppresses channel swelling and electrode delamination during electrolyte injection, while simultaneously improving charge transfer and ionic transport within the OMIEC: DtFDA-based hybrid system. Consequently, the weight ratio of OMIECs to DtFDA was systematically optimized, varying from 4:1 to 5:1, 6:1, 7:1, and 8:1, to evaluate the performance of vOECTs based on different OMIECs (Figure 4b–e; Figures S12–S15), with results summarized in Table S1. All vOECTs fabricated using TdQ-DT2FBT exhibited markedly enhanced ambipolar characteristics compared to their planar counterparts. Taking the TdQ-DT2FBT-based vOECT with the optimal DtFDA ratio of 6:1 as an example, in p-type operation, the device delivers a drain current of 35.87 mA at $V_{DS} = -0.6$ V and $V_G = -1.0$ V, with a maximum g_m value of 211.50 ± 3.04 mS. In n-type mode, the output current reaches

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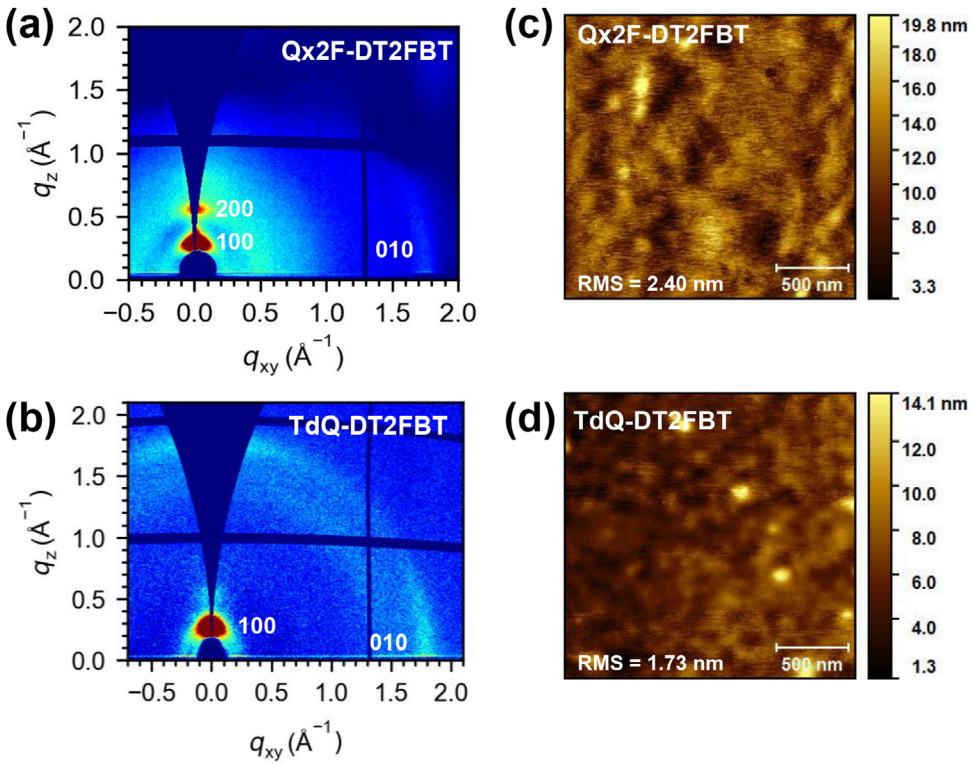


FIGURE 5 | (a,b) 2D-GIWAXS images of Qx2F-DT2FBT and TdQ-DT2FBT films. (c,d) AFM height images of Qx2F-DT2FBT and TdQ-DT2FBT films.

10.97 mA ($V_{DS} = 0.6$ V, $V_G = 1.0$ V), with a g_m value of 79.77 ± 0.76 mS. Importantly, compared to the corresponding planar OECT, the vOECT exhibits a several-fold increase in g_m and achieves a significantly improved current on/off ratio of approximately 10^6.

To further evaluate the response time of the polymer—a critical property for applications such as sensors and complementary circuits, we conducted transient characterization tests (Figure 4f,g; Figures S16 and S17). We found that, upon incorporation of DtFDA, the blend films exhibited significantly improved response times compared to the pristine OMIEC materials. Notably, the TdQ-DT2FBT:DtFDA (6:1) film demonstrated the fastest switching kinetics during both the on and off processes, achieving τ_{on}/τ_{off} values of 14.33/6.70 ms in p-type and 37.25/13.34 ms in n-type modes, respectively. Moreover, the device retained excellent stability after 1 000 s of continuous switching cycles: the p-type mode maintained 93.20% of its initial current, while the n-type mode preserved 97.80% of the original current level (Figure 4h,i; Figures S18 and S19). In contrast, although the Qx2F-DT2FBT:DtFDA (6:1) film also displayed ambipolar behavior, its g_m, current on/off ratio, and switching speed in both p-type and n-type operating regimes were markedly lower than those observed for the TdQ-DT2FBT-based devices (Figure 4j–l; Figure S20).

2.5 | Film Microstructures

The molecular packing structure and thin-film morphology of these materials were systematically investigated using grazing-incidence wide-angle X-ray scattering (GIWAXS) and atomic force microscopy (AFM). As shown in the 2D GIWAXS patterns (Figure 5a,b) and their corresponding line-cut profiles (Figure S21), Qx2F-DT2FBT exhibits (100) and (200) diffraction peaks in the out-of-plane (oop) direction and a (010) diffraction peak in the in-plane (ip) direction, whereas TdQ-DT2FBT displays only the (100) peak in the oop direction but shows (010) diffraction peaks in both oop and ip directions. These results indicate that Qx2F-DT2FBT adopts an edge-on dominant molecular packing orientation, while TdQ-DT2FBT exhibits a mixed face-on and edge-on packing structure. Subsequently, we calculated the lamellar stacking distance, π–π stacking distance, and crystal coherence length (CCL) for both polymers (Table S2). Qx2F-DT2FBT exhibits a smaller lamellar stacking distance (21.66 Å) and a larger CCL (93.46 Å) compared to TdQ-DT2FBT, which has a lamellar stacking distance of 23.70 Å and a CCL of 85.61 Å. These findings demonstrate that Qx2F-DT2FBT possesses more compact lamellar stacking and enhanced crystallinity relative to TdQ-DT2FBT, which is consistent with the contact angle measurements. However, the relatively lower crystallinity of TdQ-DT2FBT may facilitate ion penetration, which could account for its elevated volumetric capacitance C^* as verified by EIS data. Conversely, the high crystallinity of Qx2F-DT2FBT may favor hole transport while potentially limiting ionic motion, which could reduce doping efficiency. In addition, TdQ-DT2FBT exhibits stronger π–π stacking interactions, as evidenced by a larger π–π stacking CCL of 52.26 Å, in comparison to 43.80 Å for Qx2F-DT2FBT. Both the mixed packing orientation and stronger π–π stacking interactions are beneficial for charge transport, leading to improved hole and electron mobilities [50–52].

The surface morphology of the polymer thin films was characterized by AFM. The Qx2F-DT2FBT film exhibits a relatively rough

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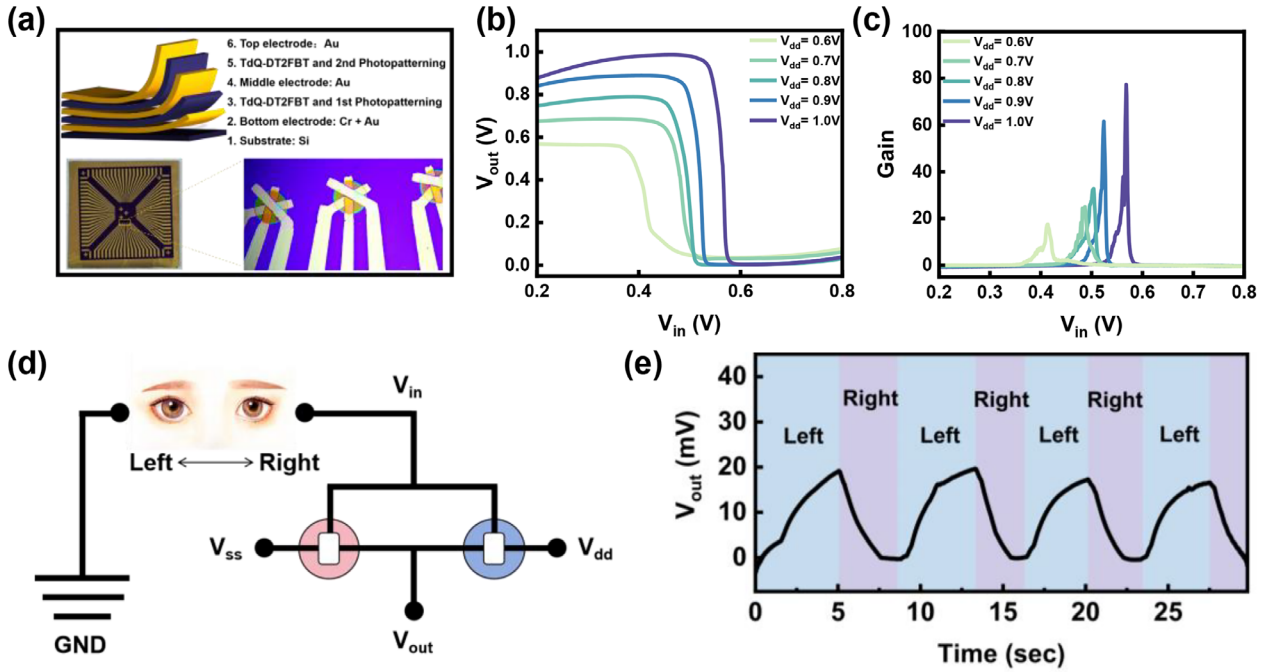


FIGURE 6 | (a) Schematic illustration of the vertical complementary inverter and optical micrographs of the fabricated device array. (b) Voltage transfer characteristics (VTCs) of the vertical complementary inverter measured at supply voltages (V_{dd}) increasing from 0.6 to 1.0 V. (c) Corresponding voltage gain as a function of input voltage (V_{in}), yielding a maximum gain of 78 V V^{-1} at V_{dd} = 1.0 V. (d) Illustration of the circuit during EOG signal monitoring. (e) The relevant EOG signal obtained via complementary inverters (approximately 20 mV potential change) during left and right eye movements.

surface, with a root-mean-square (RMS) roughness of 2.40 nm (Figure 5c; Figure S22). This pronounced surface roughness is consistent with the high crystallinity revealed in GIWAXS measurements, suggesting a stronger tendency for molecular aggregation. In contrast, the TdQ-DT2FBT film displays a significantly smoother and more homogeneous surface, with a reduced RMS roughness of 1.73 nm (Figure 5d; Figure S22). The formation of such uniform and continuous films may facilitate charge transport and reduce the probability of defect formation, thus favoring the overall performance of organic electronic devices [53, 54].

2.6 | The Ambipolar Vertical Organic Electrochemical Transistor (vOECT)-Based Complementary Inverter

Based on the outstanding ambipolar performance of TdQ-DT2FBT-based vOECTs, a vertical complementary inverter device was successfully fabricated (Figure 6a; Figure S23). Figure 6b presents the voltage transfer characteristics (V_{TC}s) measured under supply voltages (V_{dd}) ranging from 0.6 to 1.0 V. The voltage gain was determined from the slope (dV_{out}/dV_{in}) of the V_{TC} curves to evaluate the amplification capability. Notably, increasing the supply voltage to 1.0 V substantially enhances the voltage gain, achieving a maximum value of approximately 78 V V^{-1} at a voltage step size of 1 mV (Figure 6c). Benefiting from the relatively balanced p-type and n-type transport of our vertical OECTs, such high gain makes this inverter well suited for capturing weak electrophysiological signals in skin-compatible wearable electronics. OECT inverters operate at sub-1 V bias with low power and low thermal noise, outperforming traditional resistor-based amplifiers [55–59]. We therefore used this device for real-time electrooculography (EOG) recording as a material-focused proof-of-concept test. EOG signals arise from the potential difference between the cornea and retina ($V_{EOG,}$ with approximately 1.5 mV potential variations) [60]. Eye movements induce vertical or horizontal potential shifts on the surrounding skin, corresponding to upward/downward or left/right eye movements [61]. By placing two 3M-2223 electrodes on either side of the eye — one connected to the Ag/AgCl gate of the inverter and the other to ground (Figure 6d) — we observed significant and continuous fluctuations in the output voltage across three consecutive left-to-right eye movement cycles. The EOG signal was amplified more than 15 times, achieving a maximum recorded potential change of 20 mV (Figure 6e). As summarized in Table S3, our TdQ-DT2FBT-based inverter achieves comparable amplification capacity against previously reported OECT eye-tracking devices. We additionally performed 100-second continuous monitoring of EOG signals corresponding to left–middle–right eye movements. Stable signal amplification was sustained over the entire measurement period (Figure S24). With an amplification gain of roughly 8 and a maximum potential change of 12 mV, the device shows great potential for stable and reliable electrophysiological signal amplification.

3 | Conclusion

In summary, we have successfully developed a novel single-component ambipolar OMIEC, TdQ-DT2FBT, through the

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strategic integration of a quinoxaline-based backbone with the fusion of a thiadiazole unit. This molecular design effectively narrows the bandgap to 1.17 eV and lowers the LUMO energy level up to −3.78 eV, enabling efficient n-type charge transport while preserving robust p-type characteristics. As a result, TdQ-DT2FBT exhibits impressive ambipolar performance, achieving a high μC^* of 301.03 F cm^{-1} V^{-1} s^{-1} in p-type mode and 11.27 F cm^{-1} V^{-1} s^{-1} in n-type mode. We further enhanced device performance by employing vOECTs, which exhibit exceptional and relatively balanced transconductance (211.50 mS for p-type and 79.77 mS for n-type), alongside fast millisecond-scale response times and outstanding operational stability. We further successfully fabricated a vertical complementary inverter that achieves a high voltage gain of 78 V V^{-1} at a low supply voltage of 1.0 V. Finally, efficient EOG signal amplification with a gain exceeding 15 was achieved using this inverter. Overall, this study establishes quinoxaline derivatives as a viable and cost-effective class of ambipolar OMIECs, making them emerge as promising active materials for high-performance next-generation organic complementary circuits and advanced integrated bioelectronic devices.

4 | Experimental Section

4.1 | Materials

Compounds and solvents such as 3-Chloro-2-chloromethyl-1-propene, Pentaethylene glycol monomethyl ether, Sodium hydride, Tetrahydrofuran, Borane tetrahydrofuran complex solution, Hydrogen peroxide, Sodium hydroxide, 3,6-Dibromo-4,5-difluorobenzene-1,2-diamine,4,7-Dibromobenzo[c][1,2,5]thiadiazole-5, 6-diamine,Glyoxylic acid, Acetic acid, Triphenylphosphine, Diisopropyl azodicarboxylate, Hydrochloric acid, 4,7-Bis(2-3Me Sn-5-thienyl)-2,1,3-benzothiadiazole,5-Fluoro-4,7-bis(5-(trimethylstannyl)thiophen-2-yl)benzo[c][1,2,5] thiadiazole,5,6-Difluoro-4,7-bis(5-(trimethylstannyl)thiophen-2-yl)benzo[c][1,2,5] thiadiazole, Tris(dibenzylideneacetone)dipalladium(0), Tri(o-methylphenyl)phosphine, chlorobenzene were purchased from reagent suppliers for direct use and used without further purification. Supplementary information contains detailed information about molecular synthesis and characterization.

4.2 | Synthesis Methods

All the chemicals and solvents are used as received from commercial sources without further purification unless otherwise indicated. The Hydrophilic ethylene glycol (Compound 6) was synthesized according to the literature [62]. All air- and water-sensitive reactions were carried out in a nitrogen atmosphere.

4.3 | Material Characterization

Nuclear magnetic resonance (NMR) spectra, including both ^{1}H NMR and ^{13}C NMR data, were recorded on a Bruker 400 MHz AVANCE III-400 spectrometer using $CDCl_3$ as the solvent.

4.4 | Mass Spectrometry Characterization of Materials

The molecular weight of materials (excluding polymers) was determined by high-resolution point-spray ionization mass spectrometry (HR-ESI-MS) analysis using an Agilent 1100 HPLC/TOF high-performance liquid chromatography-tandem mass spectrometry system.

4.5 | Gel Permeation Chromatography (GPC) Test

The molecular weight of the polymer was determined by the PL-GPC220 high-temperature GPC system at room, chloroform as the eluent.

4.6 | Density Functional Theory (DFT) Calculation

Density functional theory (DFT) calculations employed three repeating units, with PEG chains substituted by methoxy groups for simplification. Molecular structures were optimized using the B3LYP/6-31G(d) computational model.

4.7 | Ultraviolet and Visible Spectrophotometry

Ultraviolet-visible-near-infrared absorption spectroscopy was performed using the Shimadzu UV-3600i instrument. The absorption spectra of polymer molecules were obtained in both solution and film states. The solution-state absorption spectra were measured in chloroform. The film-state absorption spectra were measured by applying a 10 mg/mL spin-coated polymer solution onto a 1 cm × 1 cm quartz plate.

4.8 | Cyclic Voltammetry (CV) Measurements

Cyclic Voltammetry measurements were conducted using a CHI760E A21731 instrument, employing a standard three-electrode configuration. A platinum foil served as the working electrode, while a platinum wire electrode and an Ag/AgCl electrode were used as the counter and reference electrodes, respectively. The measurements were carried out in either a deoxygenated, anhydrous 0.1 M tetra-n-butylammonium hexafluorophosphate ($TBAPF_6$) acetonitrile solution. The scan rate was set at 100 mV s^{-1}. The LUMO and HOMO levels are calculated by

$$E_{\mathrm{HOMO}} = -\left(4.80+E_{\mathrm{onset.ox}} - E_{\mathrm{Fc/Fc}^+}\right)\ \mathrm{eV}$$

$$E_{\mathrm{LUMO}} = -\left(4.80+E_{\mathrm{onset.red}} - E_{\mathrm{Fc/Fc}^+}\right)\ \mathrm{eV}$$

4.9 | Spectroelectrochemical Measurements

Using Instrument I-V Source-Measure as a power source, samples were spin-coated on ITO and placed in 0.1 M NaCl solution,

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voltage conditions were changed, and in situ UV–vis absorption testing was performed on a SHIMADZU UV-3600i. (The applied potential ranges from −1.0 to 1.0 V in increments of 0.1 V.)

4.10 | Electrochemical Impedance Spectroscopy (EIS)

ElS measurements were performed using a three-electrode setup: gold on glass as the working electrode (WE), platinum mesh as the counter electrode (CE), and Ag/AgCl as the reference electrode (RE). A polymer film was spin-coated onto the gold substrate and subsequently immersed in the electrolyte. EIS spectra were recorded in the frequency range from 100 kHz to 1 Hz with an AC amplitude of 10 mV (root mean square). The effective capacitance (C_{eff}) at 1 Hz is calculated using the formula $C_{eff} \sim 1/(2\pi f|Z|)$, where f is the frequency, |Z| is the magnitude of the imaginary part of the impedance, and A is the area of the gold electrode. Film thickness is measured using a step-type profilometer to determine its volume capacitance (C^*). All film thickness measurements are based on the average of three data points, with Qx2F-DT2FBT and TdQ-DT2FBT film thicknesses of 31.3 nm and 27.4 nm, respectively.

4.11 | Contact Angle Measurements

Spin-coat the polymer solution onto a silicon chip substrate (2 × 2 cm) at 3000 rpm, and measure using a droplet shape analysis system (SDC-350).

4.12 | Atomic Force Microscopy (AFM)

Atomic force microscopy (AFM) testing was performed using a Bruker Dimension Edge atomic force microscope, with scanning probe measurements conducted in tapping mode to investigate thin film morphology.

4.13 | Grazing Incidence Wide Angle X−Ray Scattering (GIWAXS)

The grazing-incidence wide-angle X-ray scattering (GIWAXS) measurements were obtained at beamline BL16B1 of Shanghai Synchrotron Radiation Facility (SSRF). The monochromatic of the light source was 1.24 Å. The data were recorded by using the 2D image plate detector of Pilatus 1 M from Dectris, Switzerland.

4.14 | Bioelectrical Signal Monitoring

To monitor real-time electrooculography (EOG) signals generated by horizontal eyeball movements and evaluate the skin compatibility of the wearable fabric devices, all experiments were conducted using the first author, Songlin Mo, as the sole participant. Prior to testing, the participant was trained on the horizontal eyeball movement protocol and was instructed to remain still and silent throughout the measurements to minimize interference from other body movements. EOG signals were recorded by placing two 3M-2223 electrodes on either side of the eye — one connected to the Ag/AgCl gate of the inverter and the other to ground. The results confirmed that the devices caused no observable irritation and were biocompatible. Therefore, the experiments were considered non-invasive and low-risk, with no ethical or safety concerns involved.

4.15 | Planar OECT Fabrication and Characterization

The [48] fabrication process for planar organic electrochemically engineered transistors (OECTs) can be summarized as follows: A mask template is applied to the silicon substrate surface using a template method. Subsequently, chromium (Cr) and gold (Au) are deposited onto the silicon substrate surface via a sputtering process at a rate of 0.1 Å/s, achieving thicknesses of 3 and 60 nm on the silicon substrate surface, respectively. After removing the mask, patterned gold/chromium electrodes serving as source and drain are obtained, with channel dimensions of W/L = 500 μm/30 μm, where W denotes electrode width, L represents the inter-electrode spacing, and d indicates film thickness. A polymer material was dissolved in chloroform to prepare a 10 mg/mL solution. A 50 μL volume was spin-coated onto the patterned substrate at 3 000 rpm. Testing employed Ag/silver chloride as the gate electrode and sodium chloride solution as the electrolyte. In this work, all reported performance values were obtained from at least three independently fabricated devices (n ≥ 3), which were prepared and characterized under identical experimental conditions. Overall, the devices exhibited reproducibility with relatively small device-to-device variation. For the planar device structure used in this study, each silicon wafer contains 12 devices. Among them, only 1–2 devices typically exhibited relatively poor performance, while the remaining devices showed relatively consistent electrical characteristics.

4.16 | Vertical OECT Fabrication and Characterization

The [48] fabrication process for planar organic electrochemically engineered transistors (OECTs) can be summarized as follows: A mask template is applied to the silicon substrate surface using a template method. Subsequently, chromium (Cr) and gold (Au) are deposited onto the silicon substrate surface via a sputtering process at a rate of 0.1 Å/s, achieving thicknesses of 3 and 60 nm on the silicon substrate surface, respectively. After removing the mask, patterned gold/chromium electrodes serving as source and drain are obtained, with channel dimensions of W/L = 500 μm/30 μm, where W denotes electrode width, L represents the inter-electrode spacing, and d indicates film thickness. A polymer material was dissolved in chloroform to prepare a 10 mg/mL solution. A 50 μL volume was spin-coated onto the patterned substrate at 3 000 rpm. Testing employed Ag/silver chloride as the gate electrode and sodium chloride solution as the electrolyte. In this work, all reported performance values were obtained from at least three independently fabricated devices (n ≥ 3), which were prepared and characterized under identical experimental conditions. Overall, the devices exhibited reproducibility with relatively small device-to-device variation. For the planar device structure used in this study, each silicon wafer contains 12 devices. Among them, only 1–2 devices

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typically exhibited relatively poor performance, while the remaining devices showed relatively consistent electrical characteristics.

4.17 | Inverters based vOECTs Fabrication

The [48] fabrication process of vertical inverters is similar to that of VOECTs. A mask template was first applied to a silicon substrate using the template method. Chromium (3 nm) and gold (60 nm) layers were subsequently deposited onto the substrate by vapor deposition. The TdQ-DT2FBT:DtFDA blend solution was spin-coated on the substrate at 3 000 rpm for 30 s to form the channel film. Subsequently, the channel film was cross-linked by exposure under 365-nm UV light for 2 min with a shadow mask, and gently wash the non-crosslinked portion with chloroform, then dried with nitrogen. A middle electrode mask template was aligned under an optical microscope, followed by the vapor deposition of a 60 nm gold layer onto the active film. Next, spin-coat the TdQ-DT2FBT:DtFDA blend solution using the same parameters. Crosslink the channel film via the same method: the channel film was cross-linked by exposure under 365-nm UV light for 2 min with a shadow mask, and gently washed the non-crosslinked portion with chloroform, then dried with nitrogen. Finally, a top electrode mask template was aligned under an optical microscope, followed by the vapor deposition of a 60 nm gold layer onto the active film.

4.18 | Statistical Analysis

For all experiments, data were obtained from at least three independently fabricated and characterized samples ($n \geq 3$). Statistical analysis and data processing were conducted using Origin 2021 (OriginLab, USA), and all results are reported as mean ± standard deviation (SD).

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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